

App Development Project Report

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1. Student Details

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About Me: I am a student at IITM, currently as software intern at IIT ROOPER

2. Project Details

Project Title: Placement Portal Application

Problem Statement:

Making an online platform which helps colleges to manage their placement activity.

Approach:

App was built with flask and vue to make a simple platform to serve the purpose.

3. AI/LLM Declaration

I have used claude code and chatgpt for 20 % into the code base writing api wrappers and frontend connections, code suggestions.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework
SQLAlchemy	Object Relational Mapper for SQLite database

Bootstrap 5	Frontend styling and responsive design
Chart.js	To visualize trends such as activity vs. time
Flask-Login	User authentication and session management
Vue 2	Frontend Modules
SQLite	Lightweight local database for storing user data
Flask Extensions	For core functionality like passwords, DB ORM

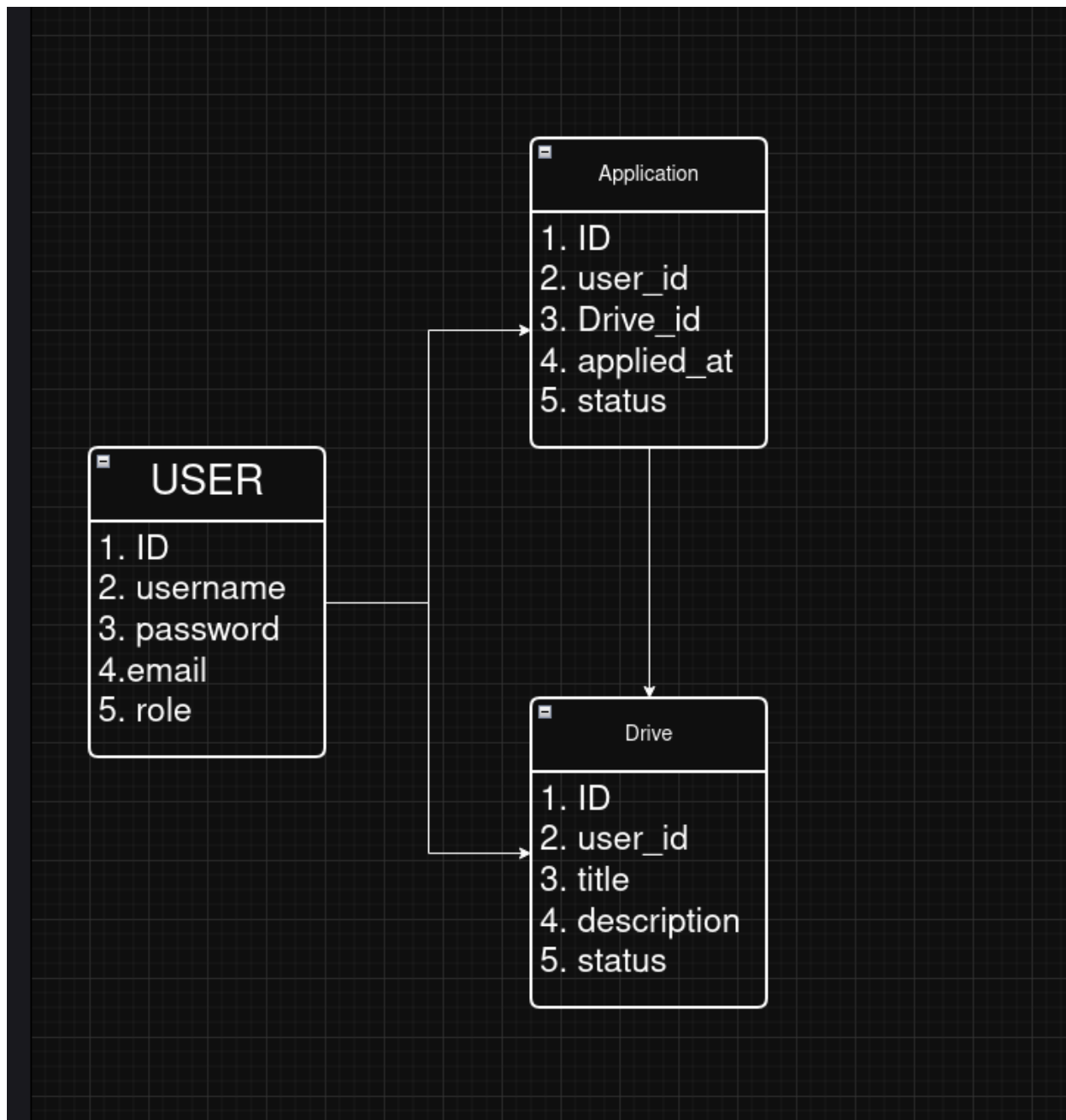
5. Database Schema / ER Diagram

Tables:

1. **User** — distributes role, controlled activation and dashboard core data model
2. **Application** — Student made application for drive
3. **Drive** — Created by company for student to apply into after verification

Relationships:

- One-to-Many → **User** → **Application**, **Company** → **Drive**, **Drive** → **Application**
- One-to-One → **Application** → **drive**



6. API Resource Endpoints

Endpoint	Method	Description
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<code>/api/admin/drives/pending</code>	GET	Pending Drives
<code>/api/admin/drives</code>	GET	List All drives
<code>/api/admin/drive/<int:drive_id>/approve</code>	PUT	Update drive status
<code>/api/admin/drive/<int:drive_id>/reject</code>	PUT	Reject given drive
<code>/api/admin/comapany/<int:company_id>/approve</code>	PUT	Approve company
<code>/api/admin/company/<int:company_id>/reject</code>	PUT	Reject Company
<code>/api/admin/students</code>	GET	Listing all student
<code>/api/admin/companies</code>	GET	List all company
<code>/api/user/<int:user_id>/block</code>	PUT	Block a user
<code>/api/user/<int:user_id>/unblock</code>	PUT	Unblock a User account
<code>/api/auth/register</code>	POST	Registration
<code>/api/auth/login</code>	POST	Login
<code>/api/company/drives</code>	POST	Creating Drive
<code>/api/company/drives</code>	GET	Listing all drives made
<code>/api/company/drive/<int:drive_id>/applications</code>	GET	Listing applications for a student id
<code>/api/company/applications</code>	GET	Every applicant which this company have made so faar
<code>/api/company/application/<int:application_id>/resume</code>	GET	Fetch resume for given application

<code>/api/student/placement-history/csv</code>	GET	Getting csv of placement history
<code>/api/student/<int:drive_id>/apply</code>	POST	Applying to a drive
<code>/api/student/resume</code>	POST	Uploading resume to platform
<code>/api/student/resume</code>	GET	Download own resume
<code>/api/user/me</code>	GET	User identity fetch

7. Architecture and Features (optional)

Architecture Overview:

- **app.py** – main Flask application entry point
- **/models** – database models using SQLAlchemy
- **/services** – core functions for operations
- **/routes** – Flask Blueprints for user and activity routes

Implemented Features:

Student registration and login

Company registration and admin approval flow

Visualization dashboard using Chart.js

Drive creation at company and to student dashboard via admin approval

Responsive UI built with Bootstrap

Secure password hashing and user session handling

Student application to Drives

Resume Upload section

Additional Features:

- Export placement history logs as CSV
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8. Video Presentation

<https://drive.google.com/file/d/1GMjya1kd5Pjf8O9n4BfzP-cQm8smDtlq/view?usp=sharing>

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